

CRIMEINSIGHT: CRIME DATA PIPELINE AND BUSINESS INTELLIGENCE SYSTEM

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ABSTRACT

CrimeInsight is a data engineering and business intelligence system designed to demonstrate the complete lifecycle of processing, storing, transforming, and analysing large-scale crime data. The system collects crime records from multiple source systems such as CSV files, open datasets, and simulated streaming sources, representing real-world heterogeneous data generation. A structured data architecture is designed to support reliable data ingestion, scalable storage, data modeling, transformation, and analytical processing.

The project implements batch-based data ingestion to collect raw crime records and stores them in a raw data layer before applying validation, cleansing, deduplication, normalization, and transformation processes. The transformed data is modeled into structured datasets suitable for analytical workloads, following appropriate data modeling and storage principles. The pipeline demonstrates key data engineering considerations including data quality, schema consistency, scalability, fault tolerance, partitioning, and efficient storage.

The processed data is then loaded into an analytical storage layer where business intelligence techniques are used to identify patterns such as crime frequency, geographic distribution, temporal trends, crime categories, and changes across different periods. Interactive dashboards and reports provide meaningful insights from the processed datasets. The project therefore demonstrates major stages of the Data Engineering Lifecycle, including source systems, data generation, ingestion, storage, data modeling, transformation, processing, and analytics. CrimeInsight provides a practical implementation of modern data engineering concepts while demonstrating how well-designed data pipelines can convert raw data into reliable and actionable information.

INTRODUCTION

The rapid growth of digital reporting systems has led to the continuous generation of large and diverse volumes of crime-related data from multiple sources. These datasets may arrive in different formats and can contain missing values, duplicate records, inconsistent fields, and other data quality issues. Without a proper data engineering pipeline, processing such data efficiently and preparing it for analysis becomes difficult. Data engineering provides the architecture, tools, and processes required to collect, ingest, store, process, transform, and organize raw data into reliable information.

CrimeInsight is an end-to-end data engineering and business intelligence system designed to process crime data and demonstrate the complete data engineering lifecycle. The system uses Python to generate and prepare realistic crime records, while Apache Kafka enables continuous ingestion of crime events into the data pipeline. PostgreSQL is used for reliable structured storage of operational and processed data, and Apache Spark performs data cleaning, validation, transformation, filtering, and aggregation. Apache Airflow orchestrates the different pipeline stages and automates scheduled data processing, while dbt is used to organize analytical transformations and data models. The processed data is finally connected to Power BI to create interactive dashboards for analyzing crime categories, temporal trends, geographic distributions, and other patterns. Docker is used to provide a consistent environment for the project components. The system brings together essential data engineering concepts including source systems, data generation, data architecture, storage systems, real-time and batch ingestion, data processing, data modeling, transformations, data quality, orchestration, and business intelligence.

THANK YOU